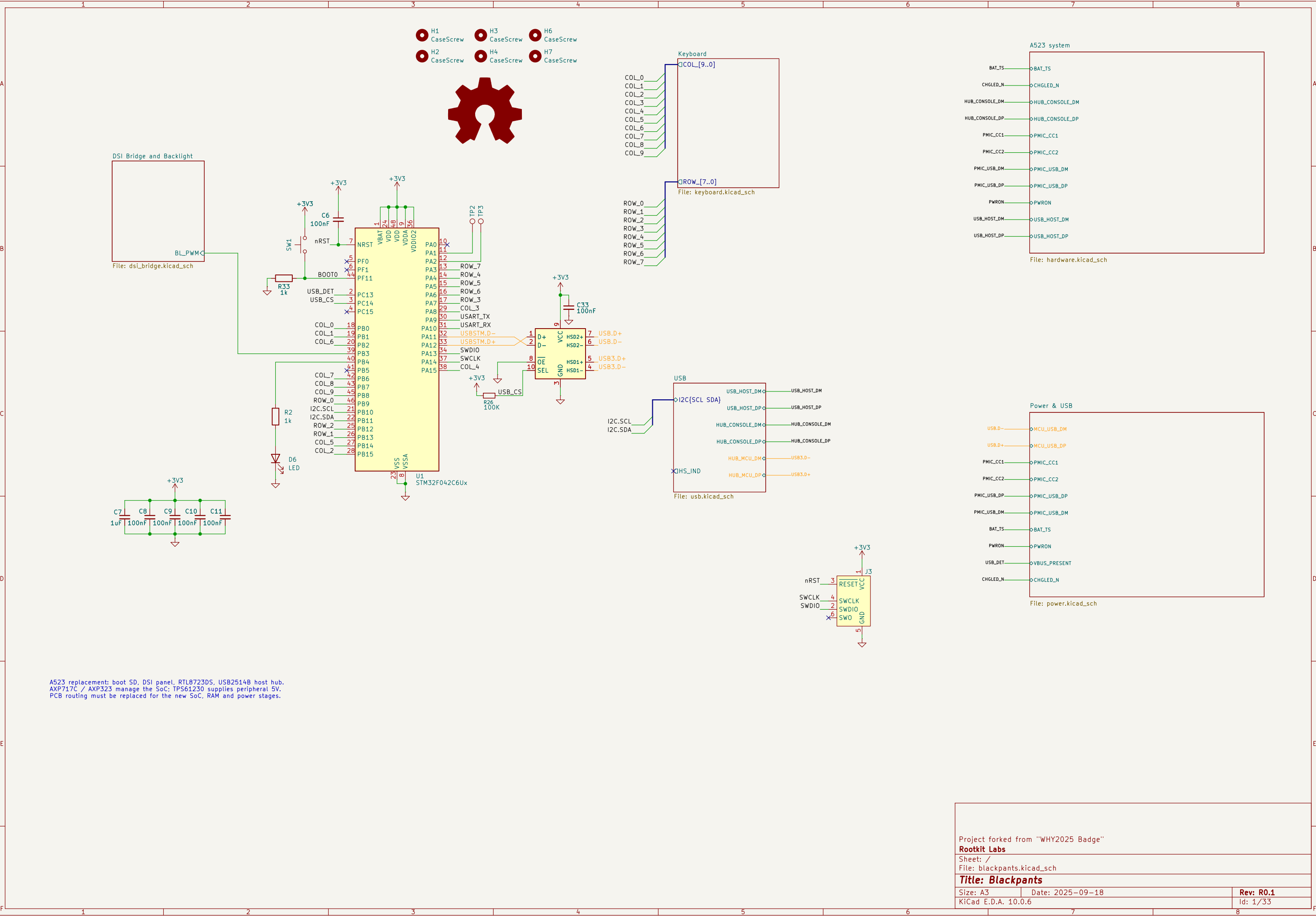
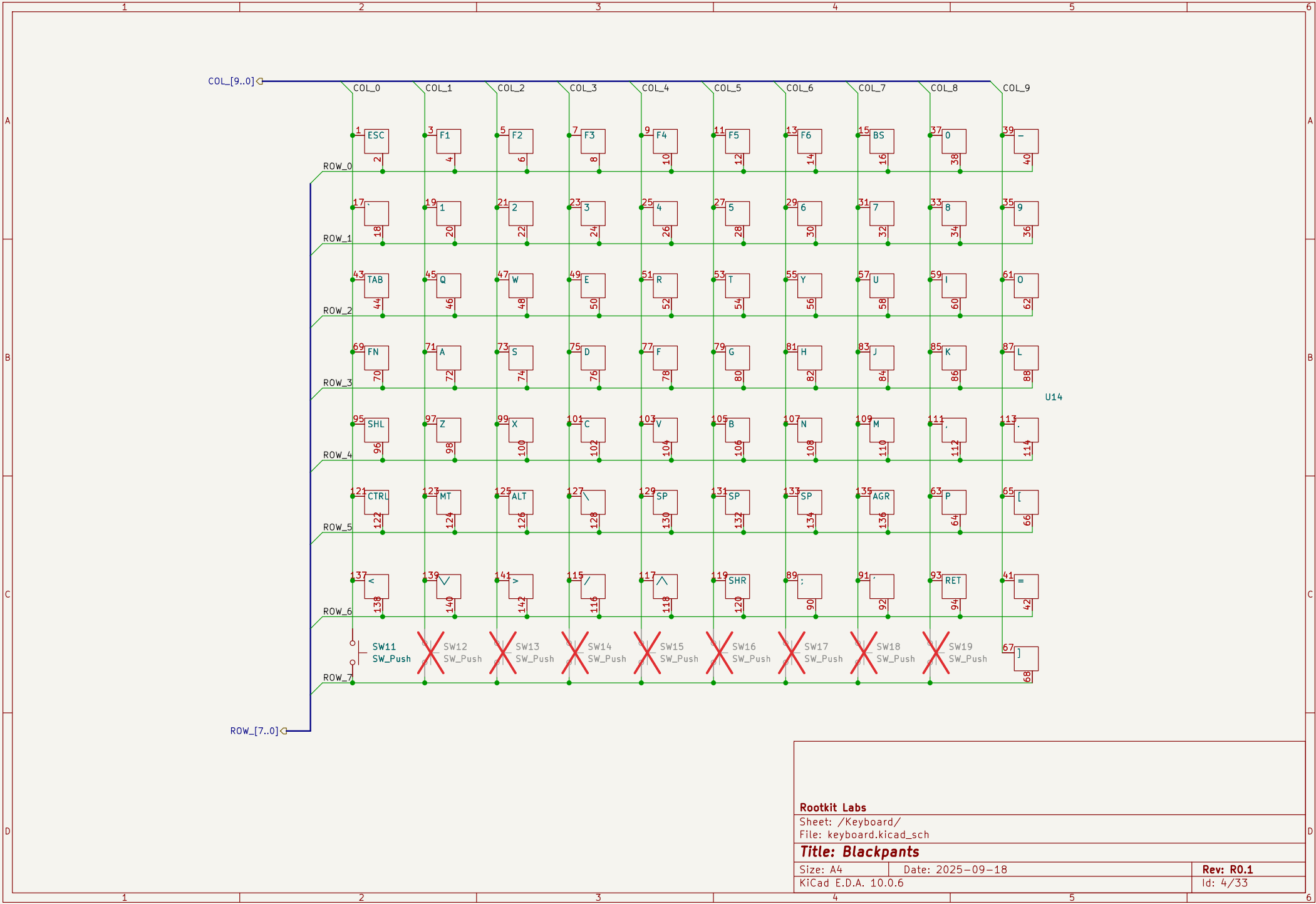
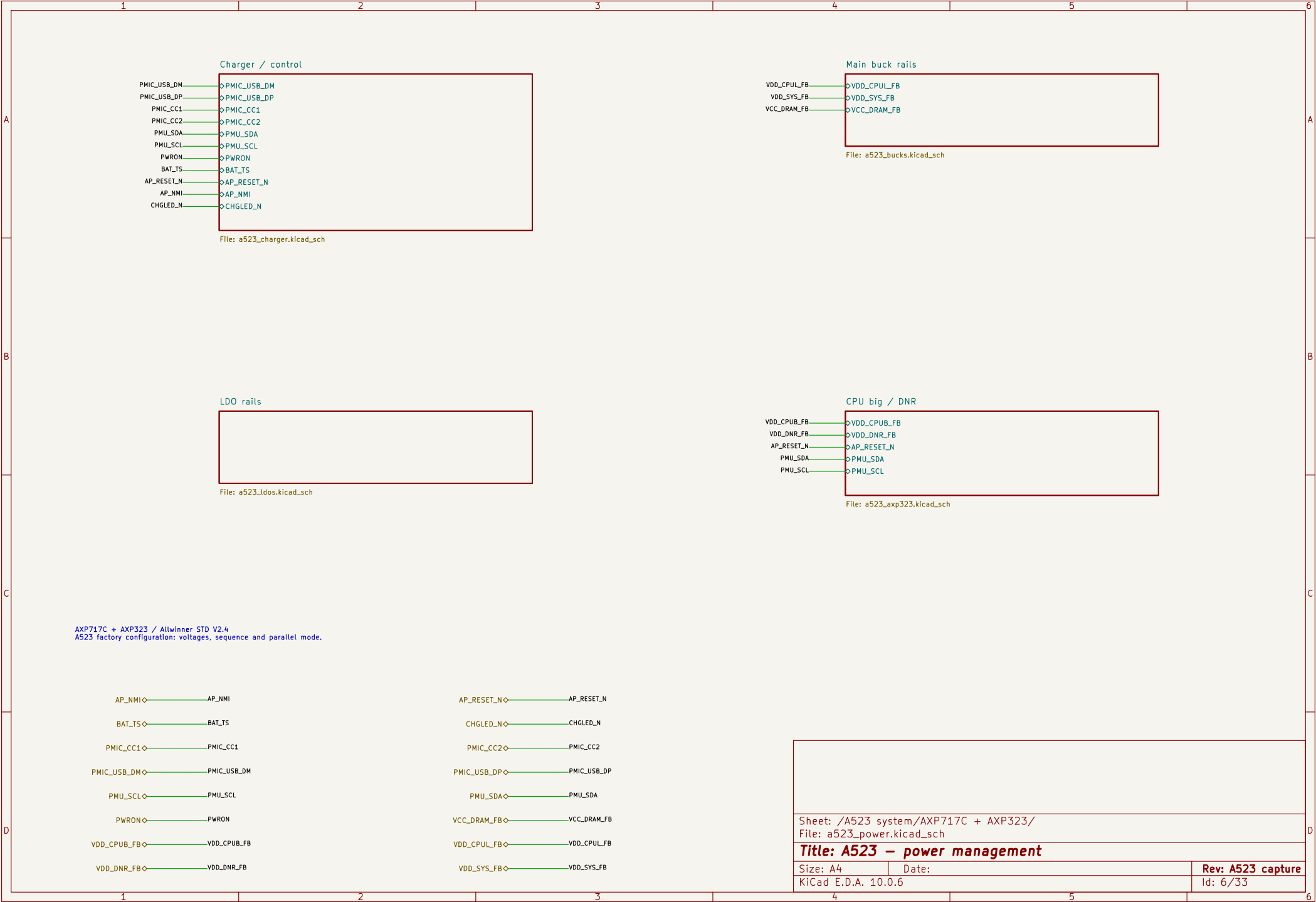
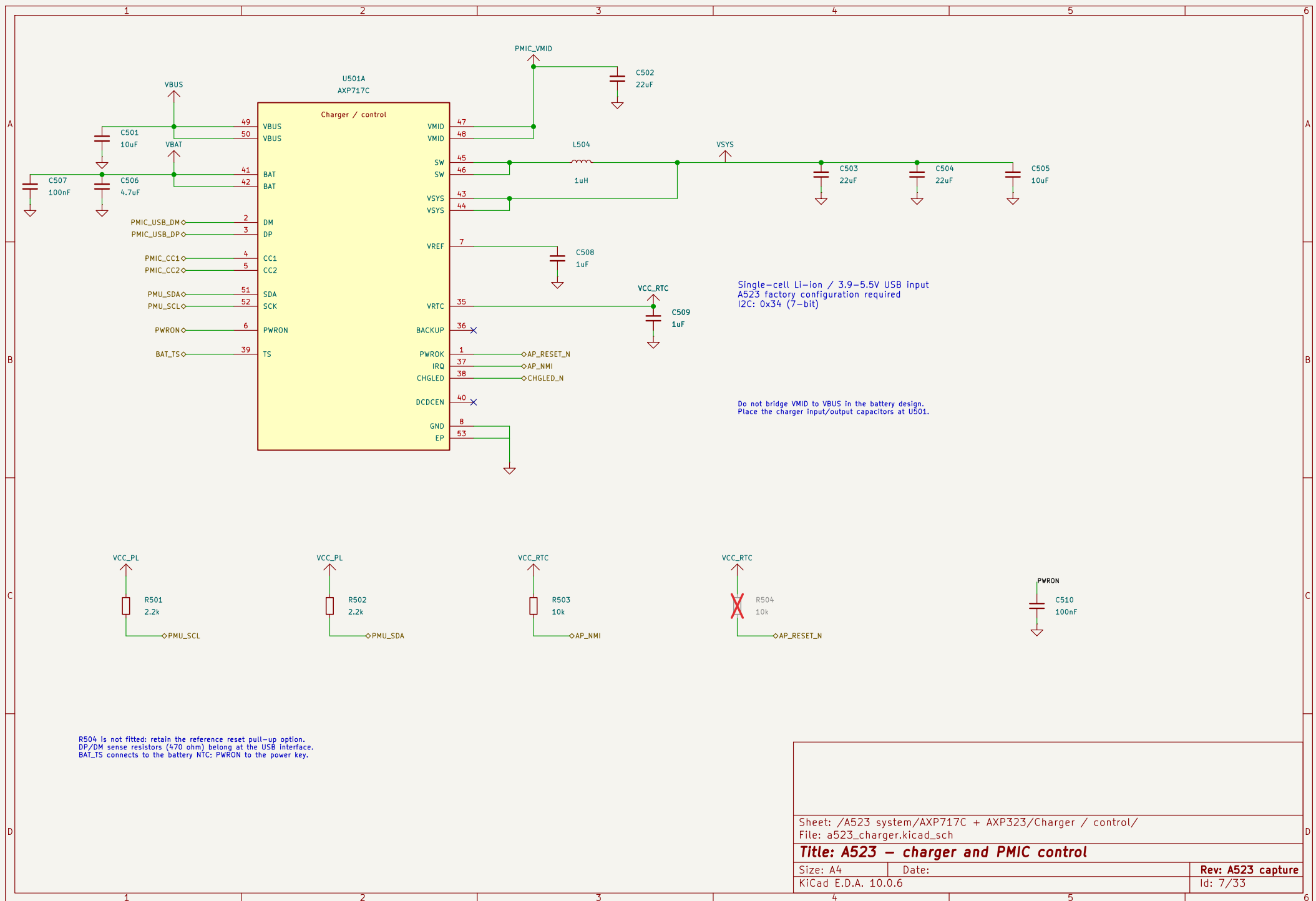
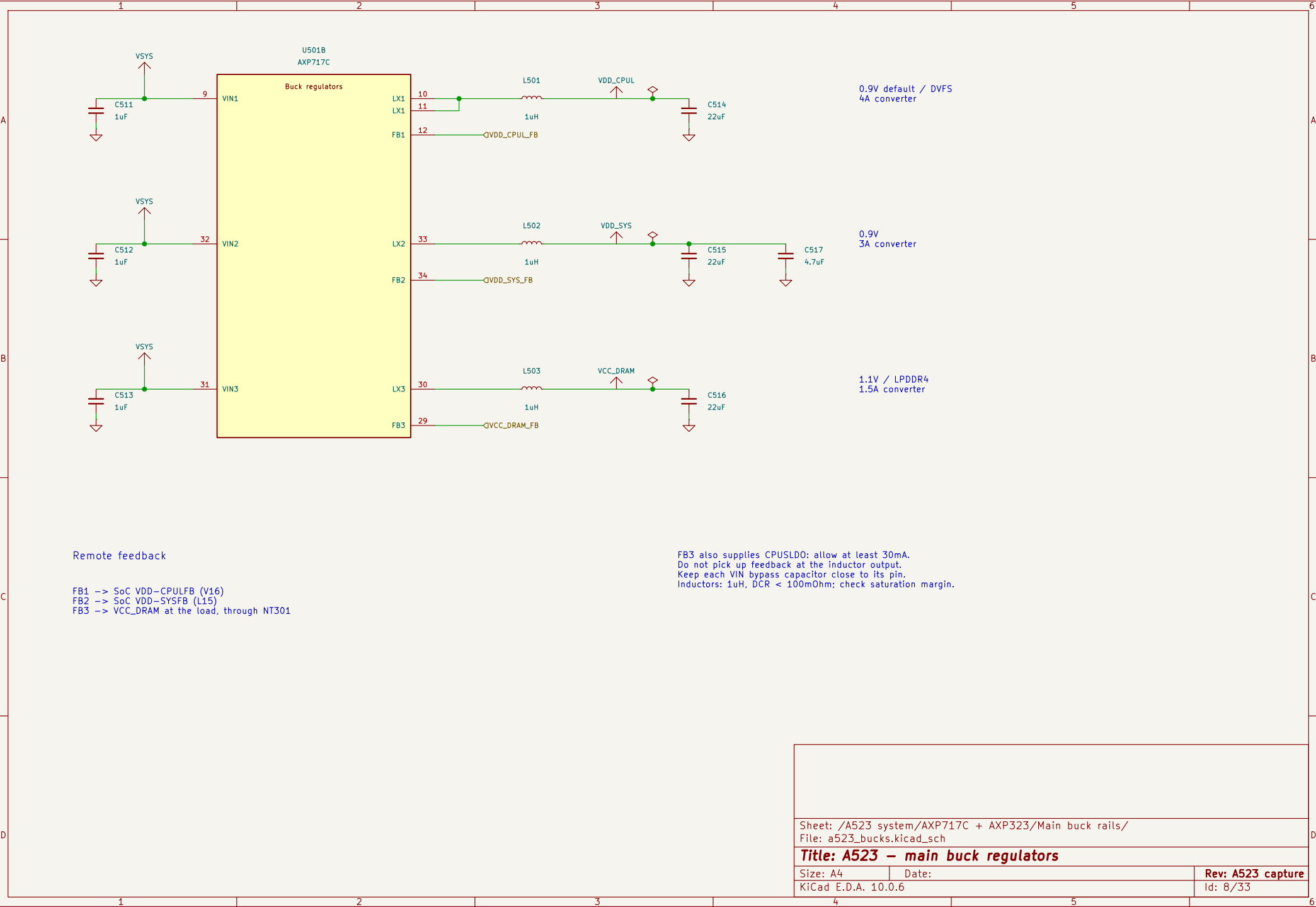


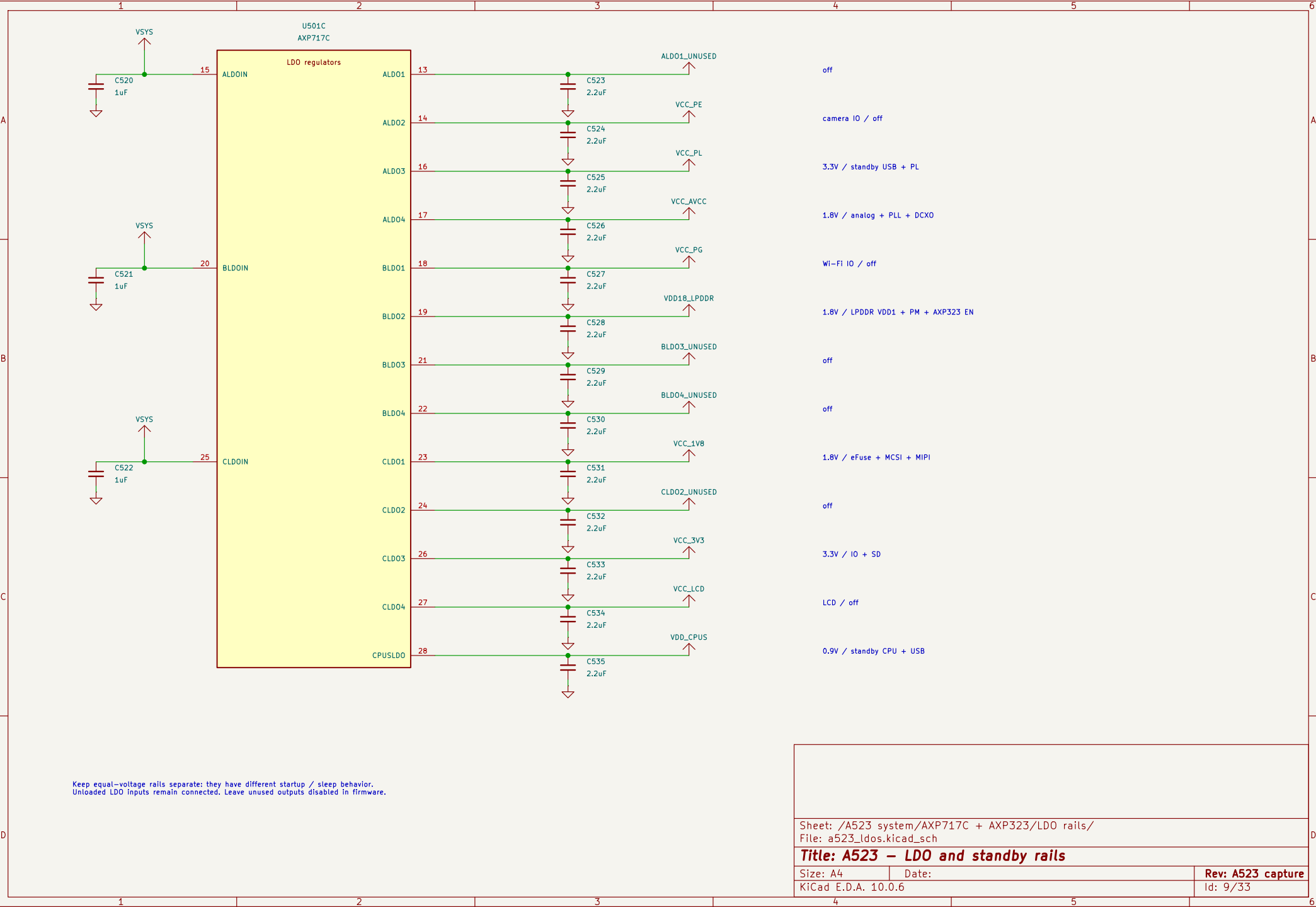


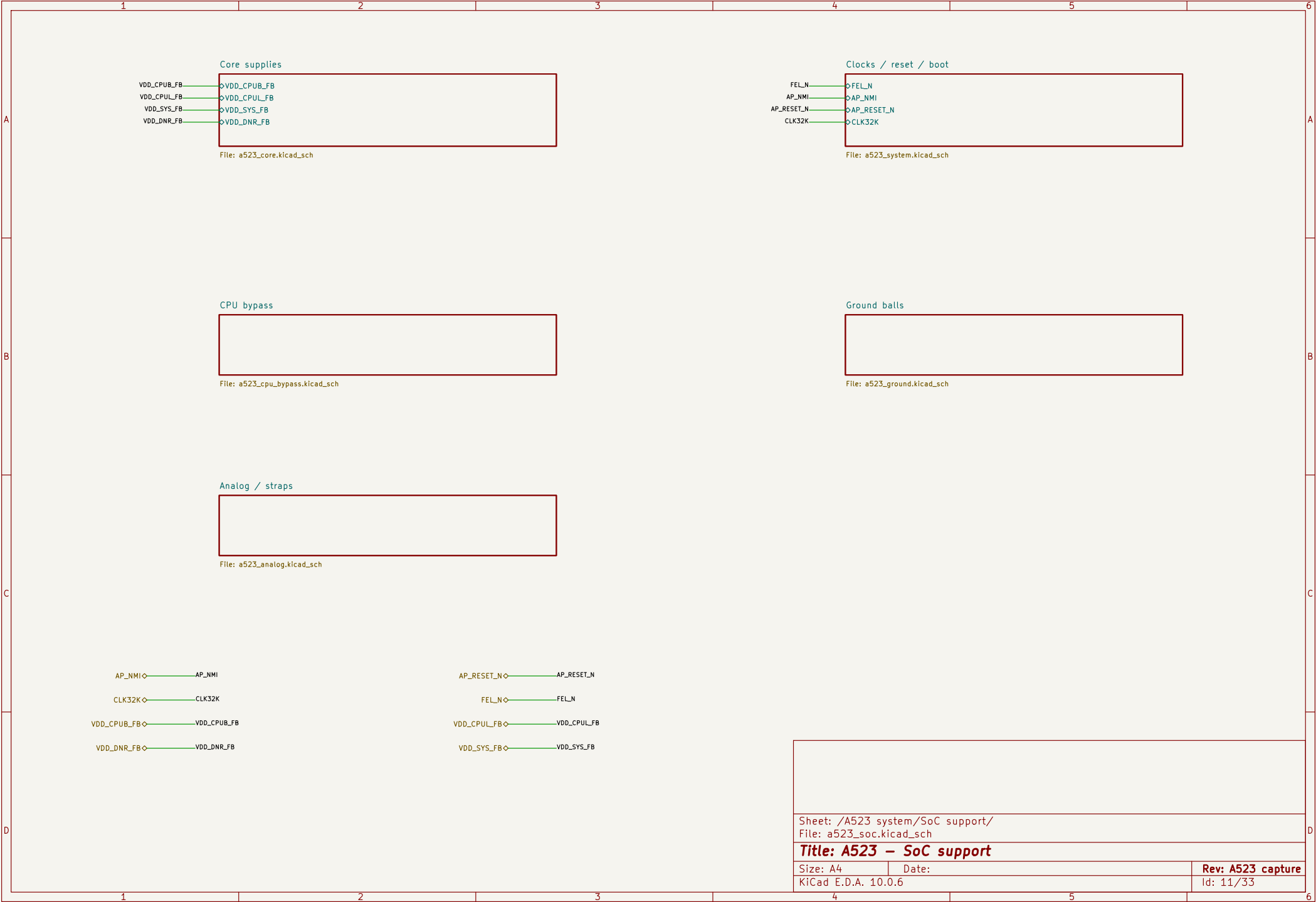




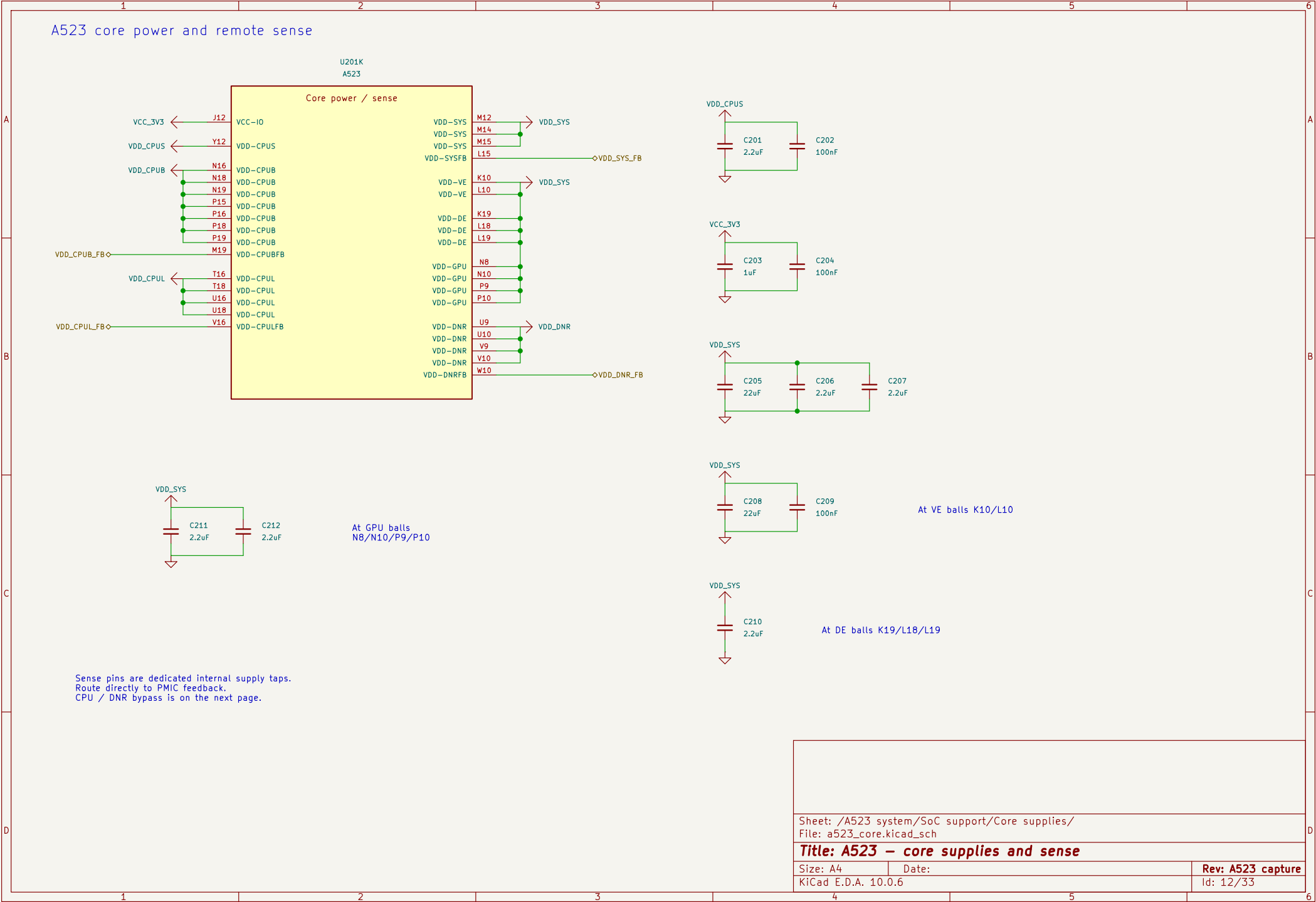




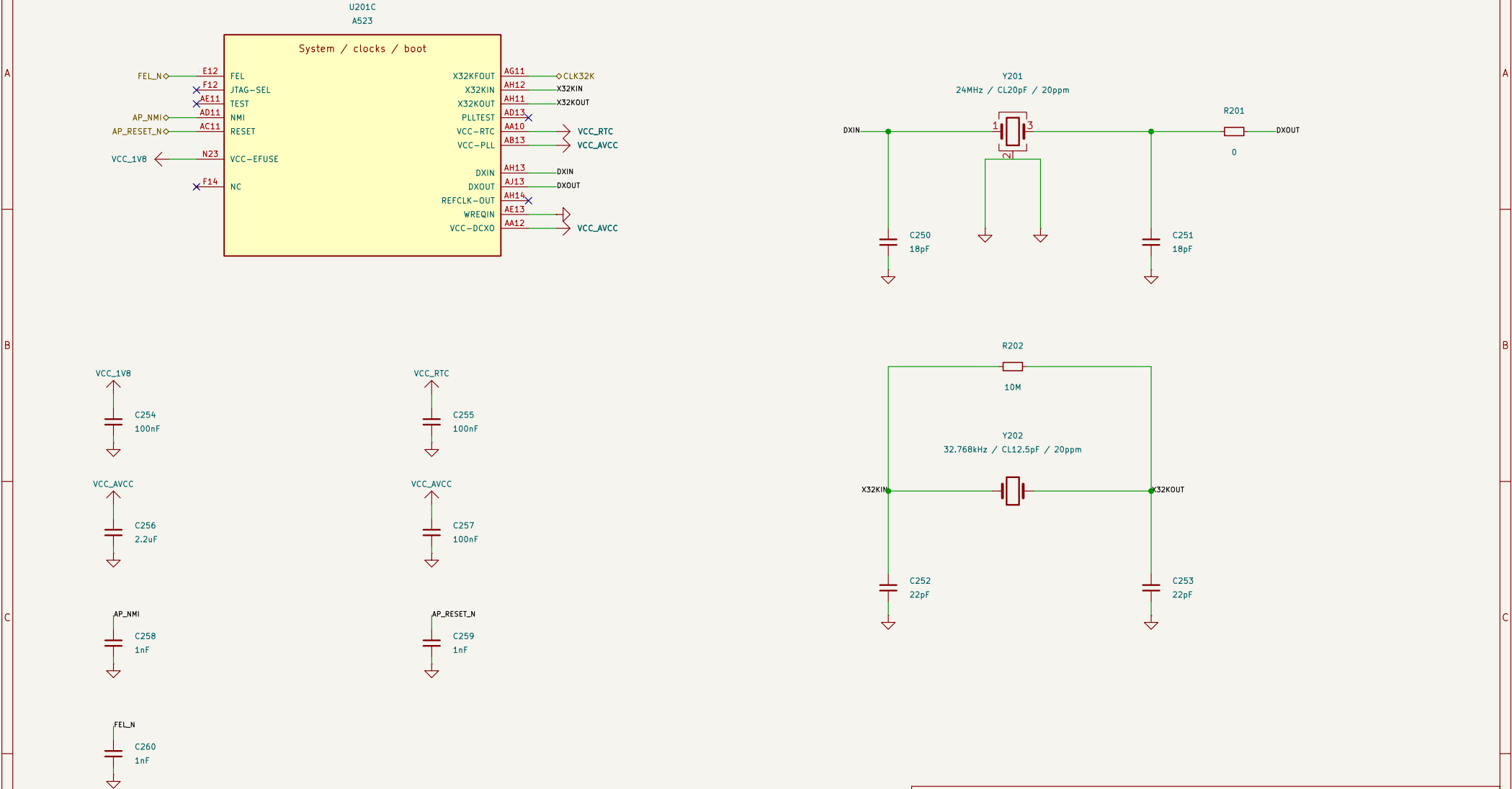




A523 core power and remote sense

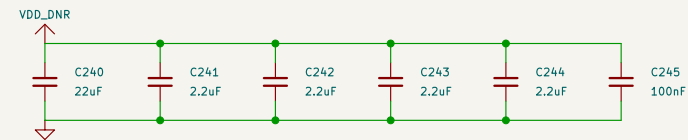
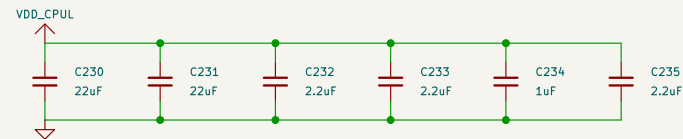
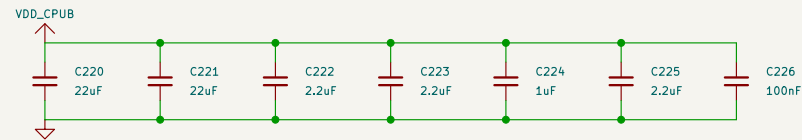


System clocks / reset / FEL



TEST / JTAG-SEL / PLLTEST / REFCLK-OUT: unconnected.
WREQIN grounded; no Allwinner Wi-Fi clock fanout.
C258/C259: 1nF NMI / reset filters.
Tune crystal capacitors for CL and PCB stray capacitance.

CPU / DNR decoupling: place at the SoC balls



Populated capacitances follow Allwinner STD / hardware guide p39.
Keep the individual capacitors at the corresponding ball clusters; bulk PMIC output capacitors are additional.
Use X5R/X7R parts with adequate effective capacitance at the operating voltage.
Feedback routing: M19 CPUB, V16 CPUL, L15 SYS, W10 DNR.

The schematic illustrates the ground ball connections for three A523 components: U201L, U201M, and U201N. Each component has a central yellow box representing its internal ground structure.

- U201L and U201M:** These components have two digital ground planes. U201L's left plane is labeled "Digital ground 1" and its right plane is labeled "Digital ground 2". U201M's left plane is labeled "Digital ground 1" and its right plane is labeled "Digital ground 2". Each digital ground plane is connected to a common ground point (indicated by a downward arrow) via a series of ground balls. The ground balls are labeled with component identifiers (e.g., A1, A2, A29, B29, C3, C6, C8, C10, C24, C25, C26, C27, D12, E3, E24, E27, G3, G6, G7, G23, G24, G27, H9, H11, J3, J10, J22, J26, K9, K12, K15, K16, K18, L3, L6, L7, L9, L13, L16, L21, L23, M9, M10, M16, M21, N3, N7, N12, N13, N15, N21, P11, P13, R3, R6, R7, R15, R16, R23, R26, T9, T10, T12, T13, T19, U12, U15, U19, U20, U26, U28, V12, V13, V15, V18, V19, W1, W8, W9, W12, W14, W25, W29, AA6, AA7, AA11, AA17, AA18, AA19, AA20, AA25, AA28, AC13, AC21, AC24, AC27, AC29, AD7, AD23, AD28, AE15, AE17, AE19, AF11, AF13, AF28, AG5, AG7, AG12, AG13, AG14, AH3, AH10, AH16, AH22, AH25, AJ1, AJ6, AJ11, AJ18, AJ20, AJ24, AJ27, AJ29).
- U201N:** This component has an analog ground plane labeled "Analog ground". It is connected to a common ground point (indicated by a downward arrow) via a series of ground balls. The ground balls are labeled with component identifiers (e.g., A14, C16, C18, C20, C22, D14, F16, F18, F20, F22, H13, H16, H19, J13, J16, J18, J19).

124 digital GND balls and 17 AVSS balls.
Audio AGND is on the analog page.
Use a continuous ground reference and nearby vias.
Keep high-speed return paths continuous.

Sheet: /A523 system/SoC support/Ground balls/ File: a523_ground.kicad_sch		
Title: A523 – digital and analog ground balls		
Size: A4	Date:	Rev: A523 capture
KiCad E.D.A. 10.0.6		Id: 15/33

The schematic illustrates the ground ball connections for three A523 components: U201L, U201M, and U201N. Each component is represented by a yellow rectangular block with its internal ground planes labeled 'Digital ground 1', 'Digital ground 2', and 'Analog ground' respectively.

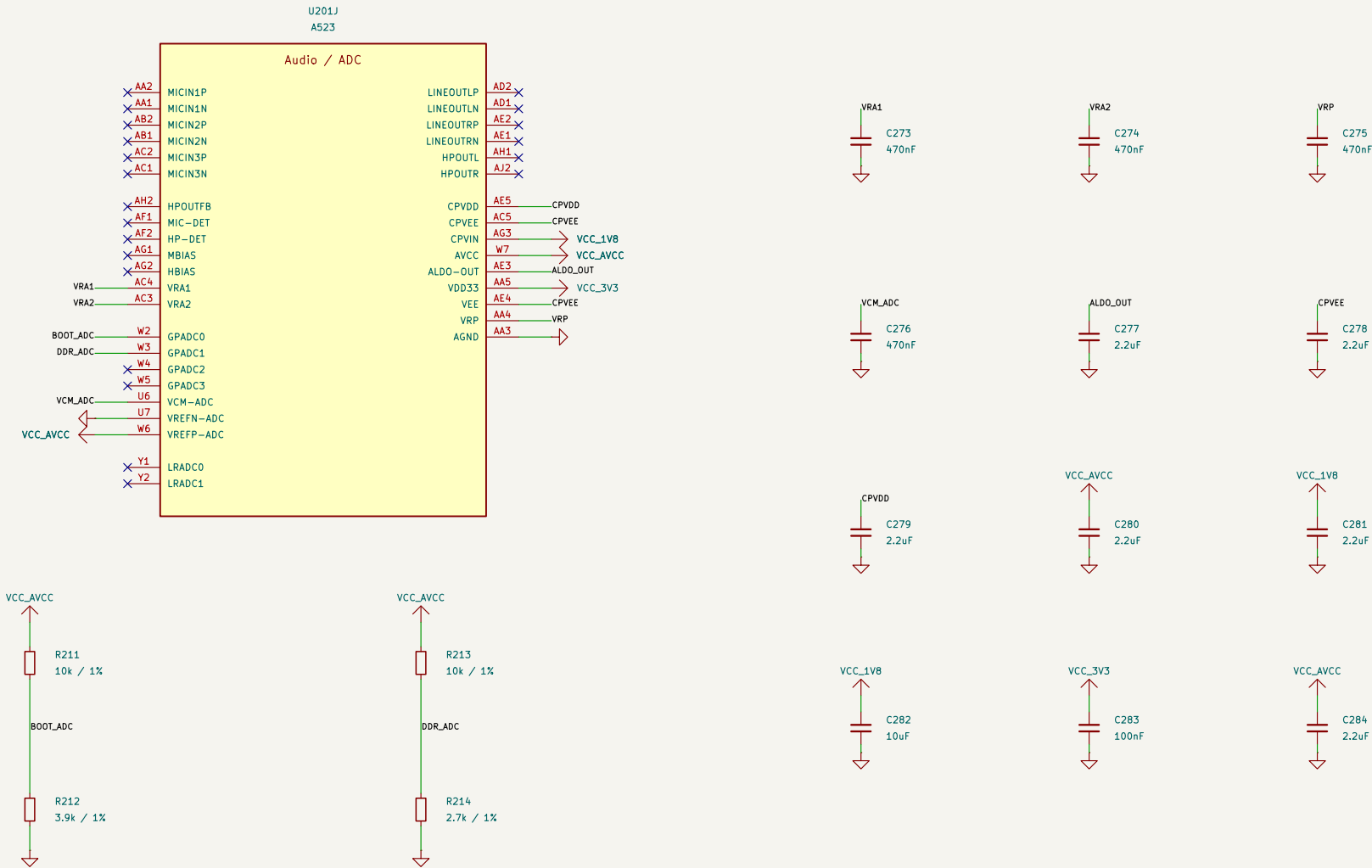
- U201L A523 (Digital ground 1):** This block has two vertical columns of pins. The left column contains 24 pins, all labeled 'GND'. The right column contains 24 pins, all labeled 'GND'. The pins are labeled with component identifiers: A1, A2, A29, B29, C3, C6, C8, C10, C24, C25, C26, C27, D12, E3, E24, E27, G3, G6, G7, G23, G24, G27, H9, H11, J3, J10, J22, J26, K9, K12, K15, K16, K18, L3, L6, L7, L9, L13, L16, L21, L23, M9, M10, M16, M21, N3, N7, N12, N13, N15, N21, P11, P13, R3, R6, R7, R15, R16, R23, R26, T9, T10.
- U201M A523 (Digital ground 2):** This block has two vertical columns of pins. The left column contains 24 pins, all labeled 'GND'. The right column contains 24 pins, all labeled 'GND'. The pins are labeled with component identifiers: T12, T13, T19, U12, U15, U19, U20, U26, U28, V12, V13, V15, V18, V19, W1, W8, W9, W12, W14, W25, W29, AA6, AA7, AA11, AA17, AA18, AA19, AA20, AA25, AA28, AC13, AC21, AC24, AC27, AC29, AD7, AD23, AD28, AE15, AE17, AE19, AF11, AF13, AF28, AG5, AG7, AG12, AG13, AG14, AH3, AH10, AH16, AH22, AH25, AJ1, AJ6, AJ11, AJ18, AJ20, AJ24, AJ27, AJ29.
- U201N A523 (Analog ground):** This block has two vertical columns of pins. The left column contains 10 pins, all labeled 'AVSS'. The right column contains 10 pins, all labeled 'AVSS'. The pins are labeled with component identifiers: A14, C16, C18, C20, C22, D14, F16, F18, F20, F22, H13, H16, H19, J13, J16, J18, J19.

Each component's ground connections are shown with green lines leading to a common ground symbol (a triangle with a horizontal line) at the bottom of the page.

124 digital GND balls and 17 AVSS balls.
Audio AGND is on the analog page.
Use a continuous ground reference and nearby vias.
Keep high-speed return paths continuous.

Sheet: /A523 system/SoC support/Ground balls/ File: a523_ground.kicad_sch		
Title: A523 – digital and analog ground balls		
Size: A4	Date:	Rev: A523 capture
KiCad E.D.A. 10.0.6		Id: 15/33

Analog support / boot and DRAM ID straps

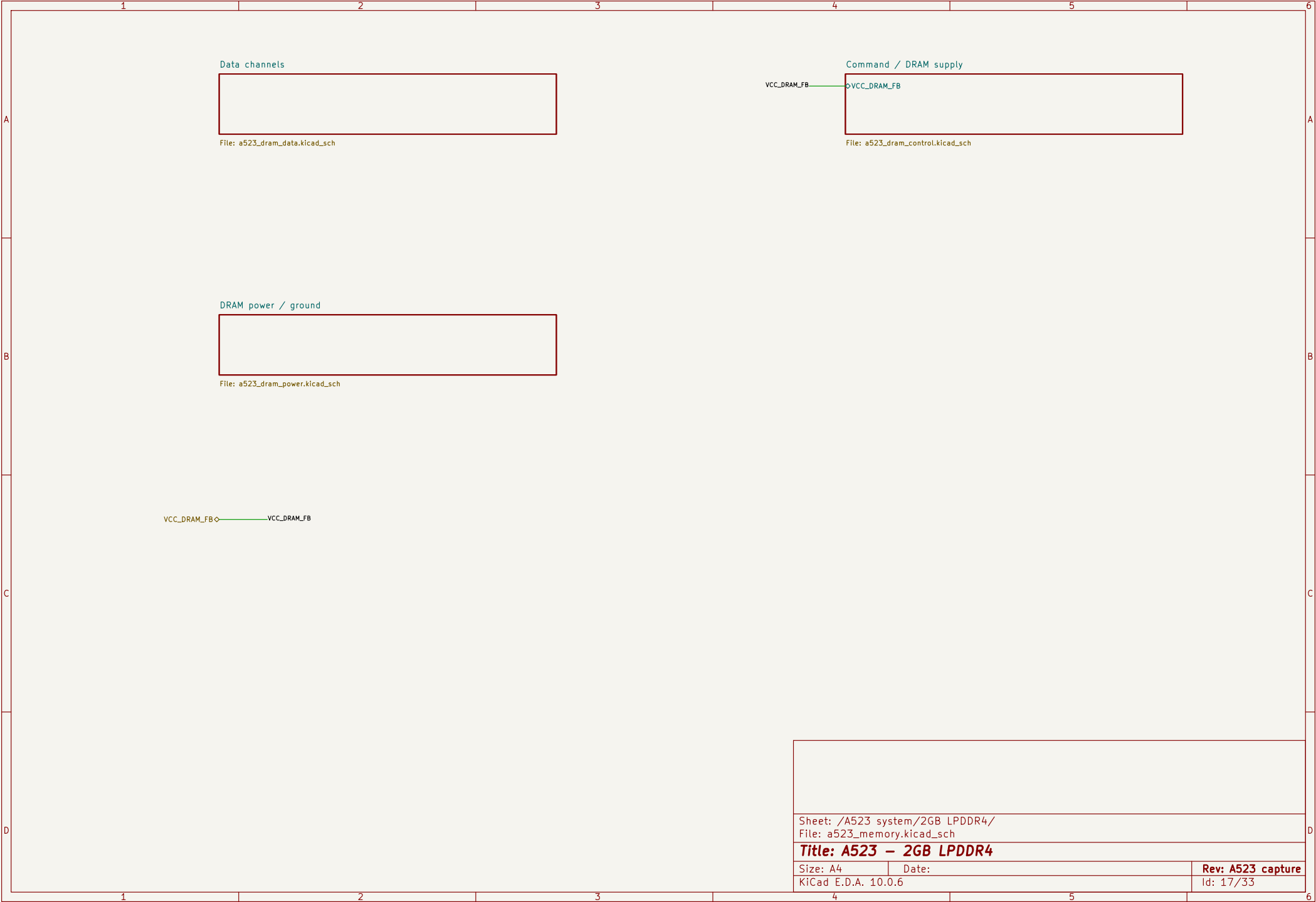


BOOT_ADC: STD 10k / 3.9k, SD boot available.
 DDR_ADC: 0.383V: PH16/PH17 = 0/0 selects set 1.
 Build set 1 for Nanya 16Gb LPDDR4.
 Unused audio / ADC IO is unconnected.
 Keep audio and headphone blocks disabled.

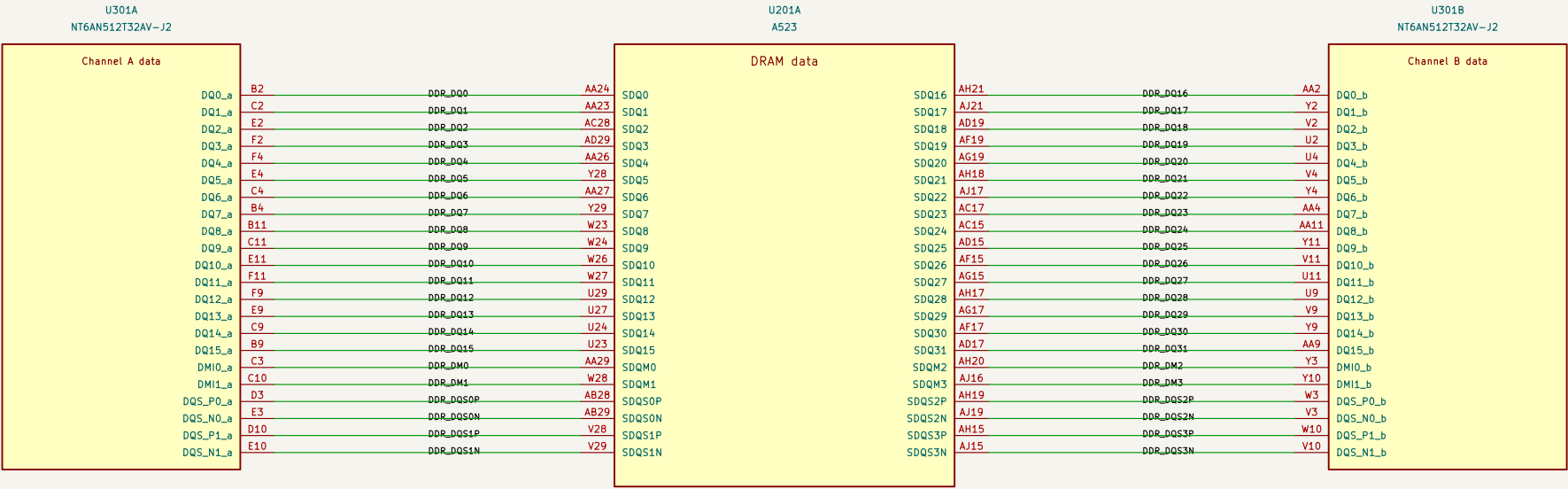
Sheet: /A523 system/SoC support/Analog / straps/
 File: a523_analog.kicad_sch

Title: A523 – analog rails and boot selection

Size: A4	Date:	Rev: A523 capture
KiCad E.D.A. 10.0.6		Id: 16/33



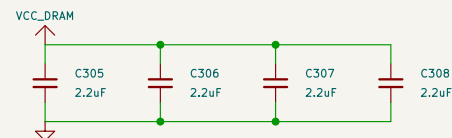
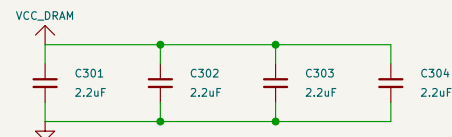
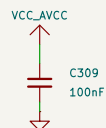
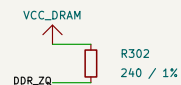
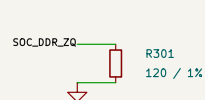
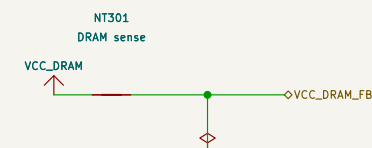
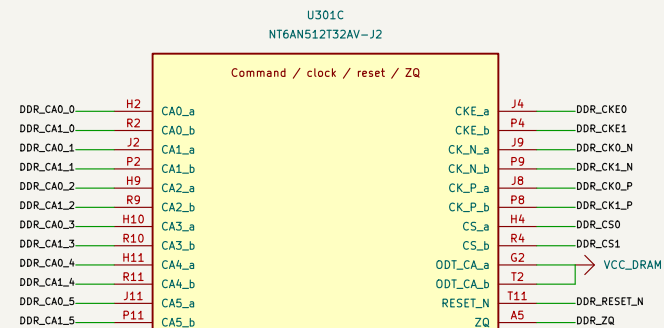
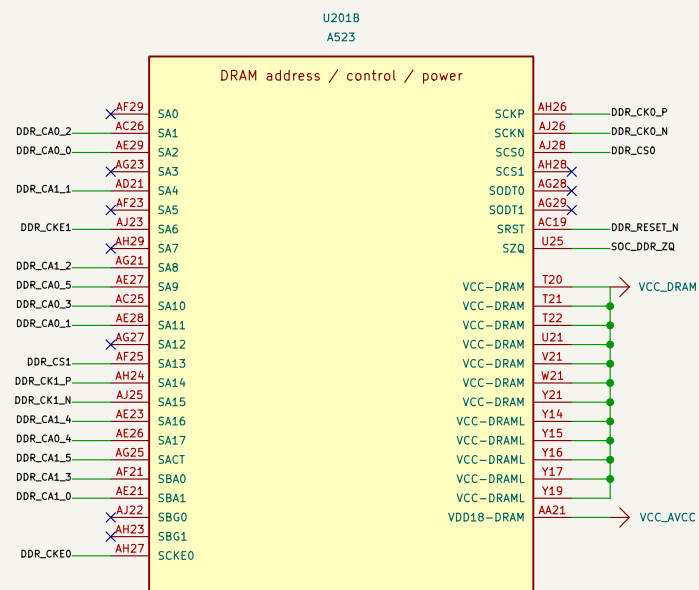
2 GB LPDDR4 / 2 x 16-bit channels / single rank per channel



Channel A: DQ[0..15], DQS/DM[0..1]
Channel B: DQ[16..31], DQS/DM[2..3]
One rank per channel. Command and power on following pages.

Vendor-qualified clock target: 1200MHz.
Use NT6AN512T32AV-J2 parameters in the DRAM firmware.
Final clock and timings require board validation.

A523 LPDDR4 command mapping / load-side feedback



NT301: load-side sense to U501 FB3; allow $\geq 30\text{mA}$.
VCC_DRAM and VCC_DRAML both use 1.1V for LPDDR4.
C301..C308 belong to the SoC DRAM ball clusters.
Unused DDR3 command functions are unconnected.

Sheet: /A523 system/2GB LPDDR4/Command / DRAM supply/
File: a523_dram_control.kicad_sch

Title: A523 – LPDDR4 command mux and SoC DRAM power

Size: A4

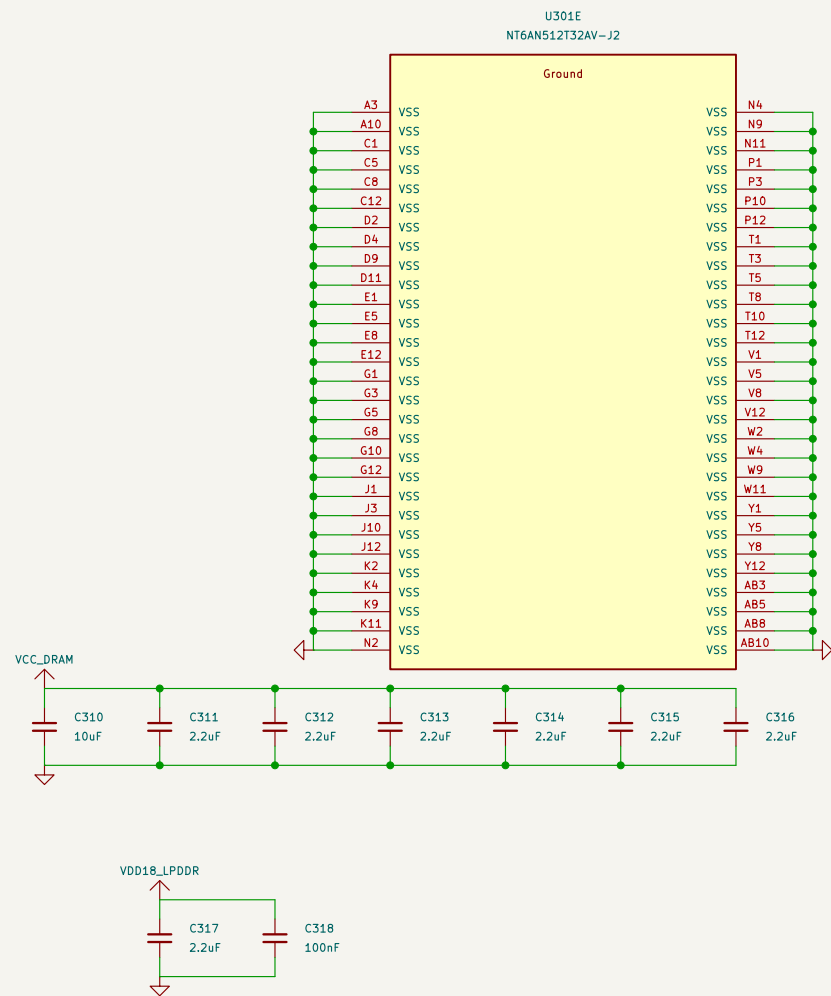
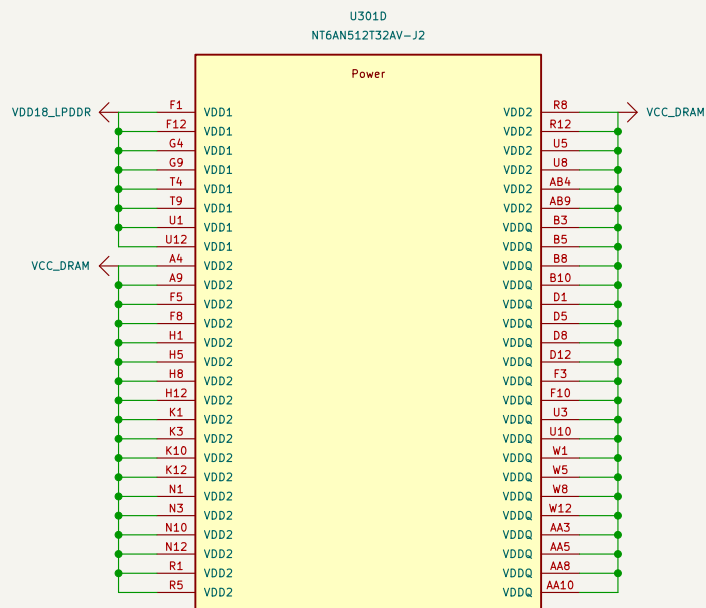
Date:

Rev: A523 capture

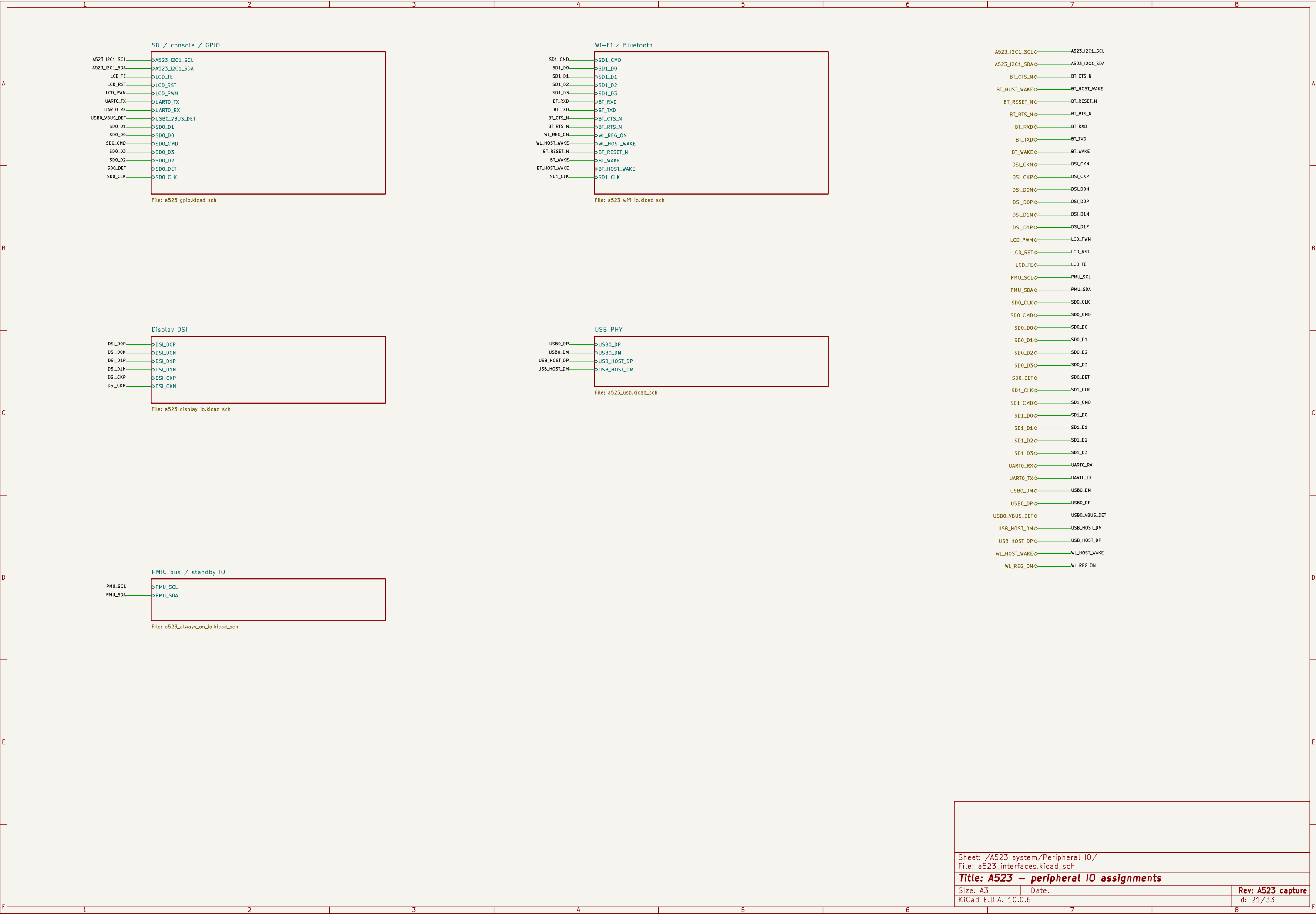
KiCad E.D.A. 10.0.6

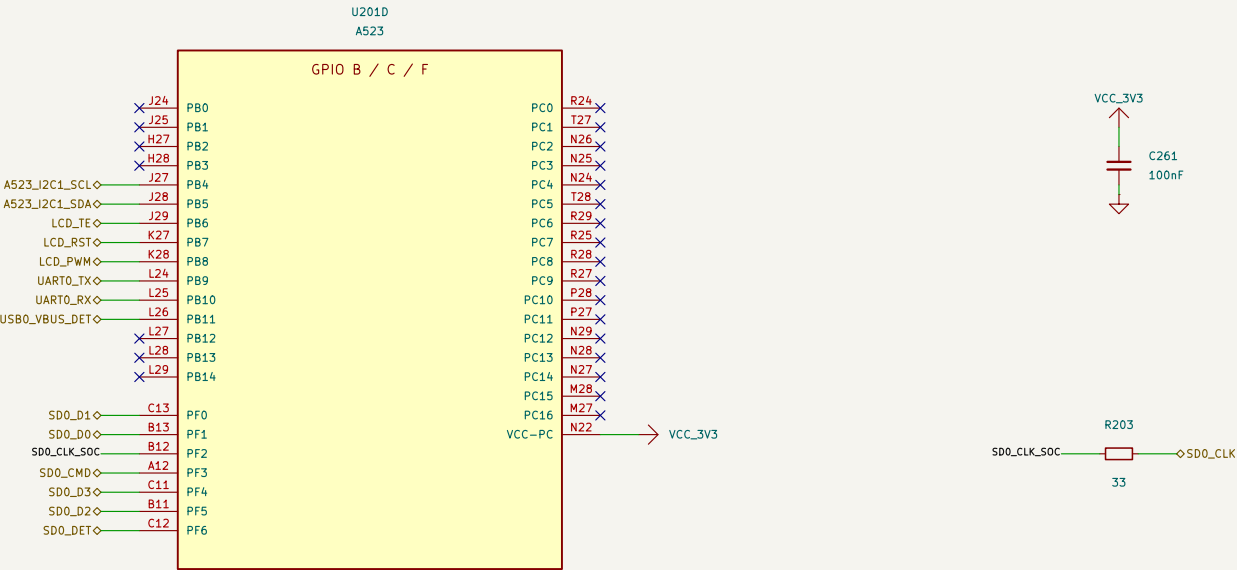
Id: 19/33

Nanya LPDDR4 supply balls and bypass



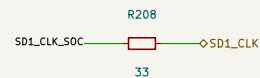
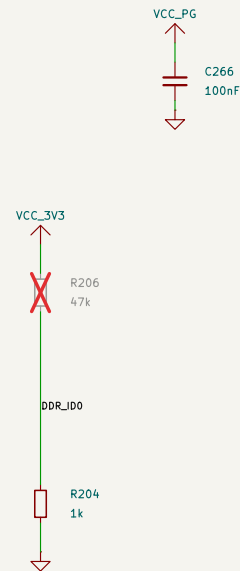
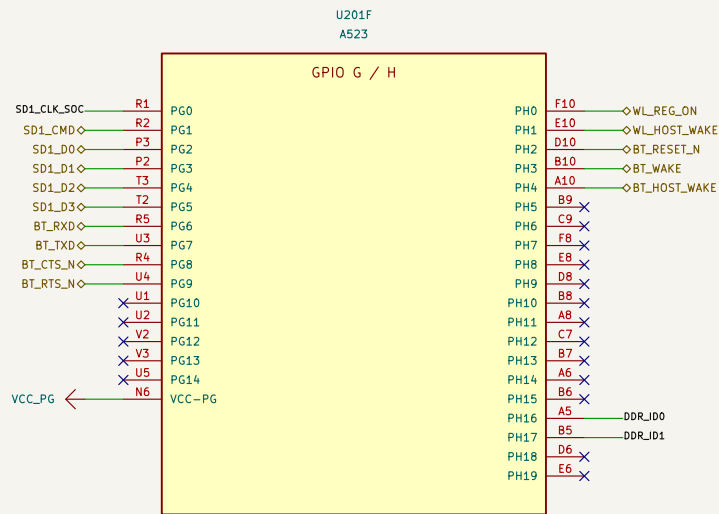
VDD1 = 1.8V BLD02. VDD2 = VDDQ = 1.1V DCDC3.
DNU pads: no tracks or vias.
Place the capacitors beside the DRAM package.
Follow Allwinner DDR constraints and the selected stack-up.





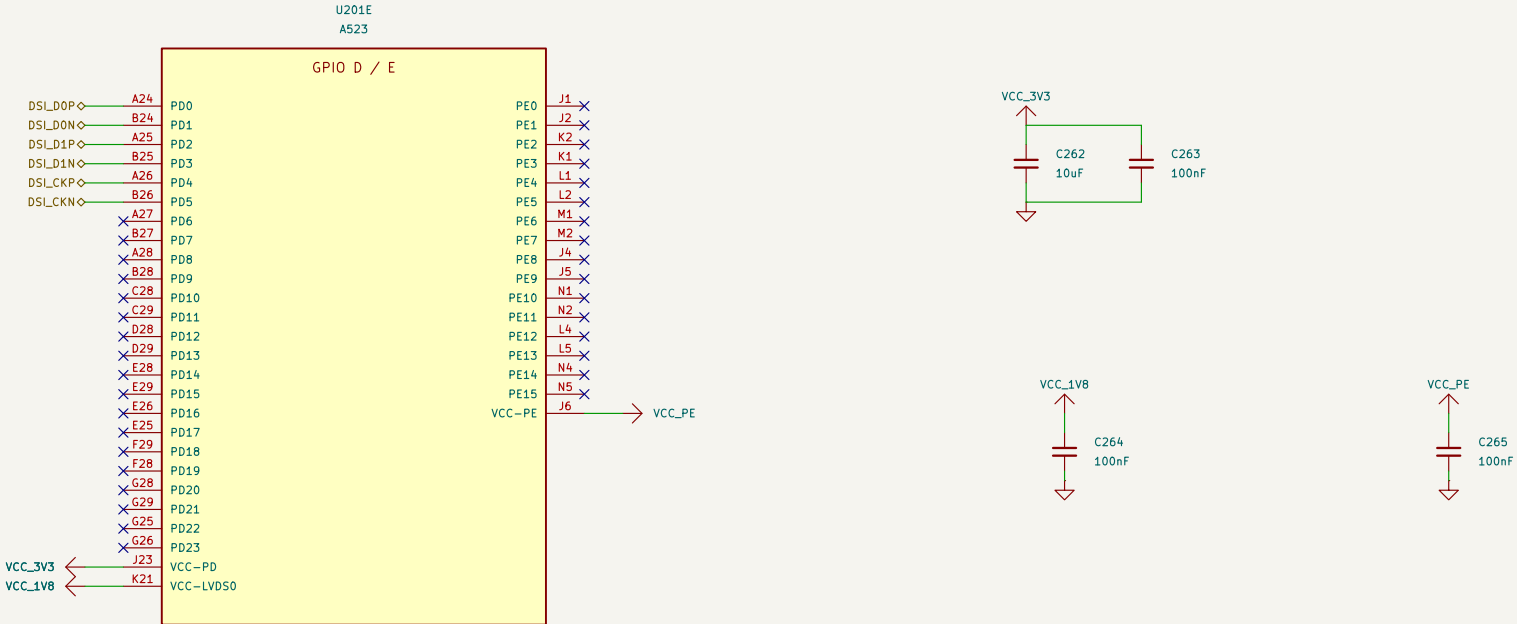
UART0: PB9 / PB10, mux 2.
SDC0: PF0..PF5, mux 2; PF6 detect.
LCD PWM: PB8 mux 6. Reset / TE: GPIO.
PB11: divided USB0 VBUS detect.
I2C1: expansion header with 3.3V pull-ups.

SDIO1 / UART1 / DDR parameter straps



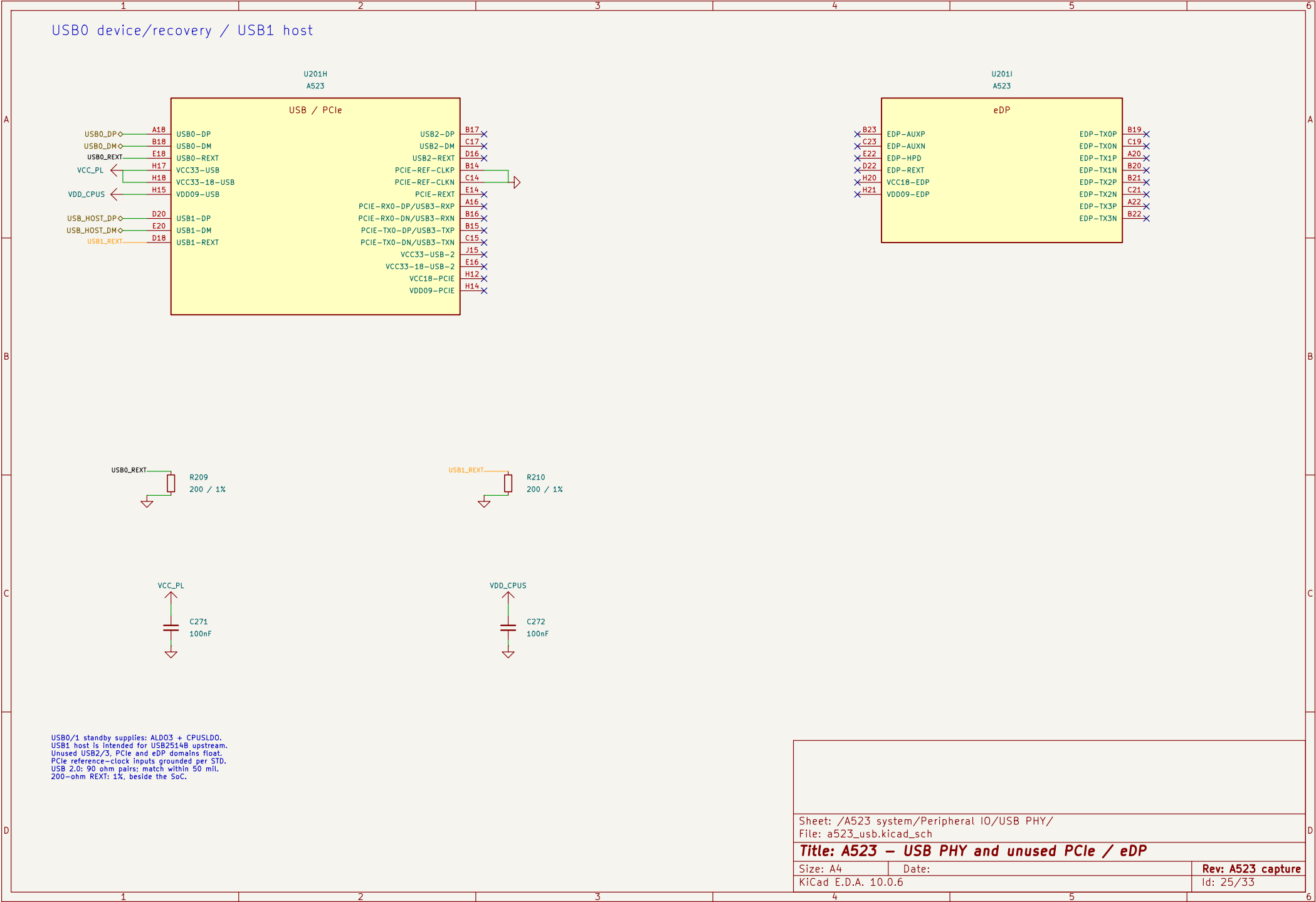
SDIO and UART use PG mux 2. PG voltage must match the module VDD_I0.
Wi-Fi/BT control pins use PH (3.3V VCC_I0); the 1.8V PM bank is not used for these controls.
PH16/PH17 = 0/0, GPADC1 selects DDR parameter set 1 (see analog page).
Firmware must populate that parameter set for NT6AN512T32AV-J2.

Two-lane DSI0 to the existing display connector

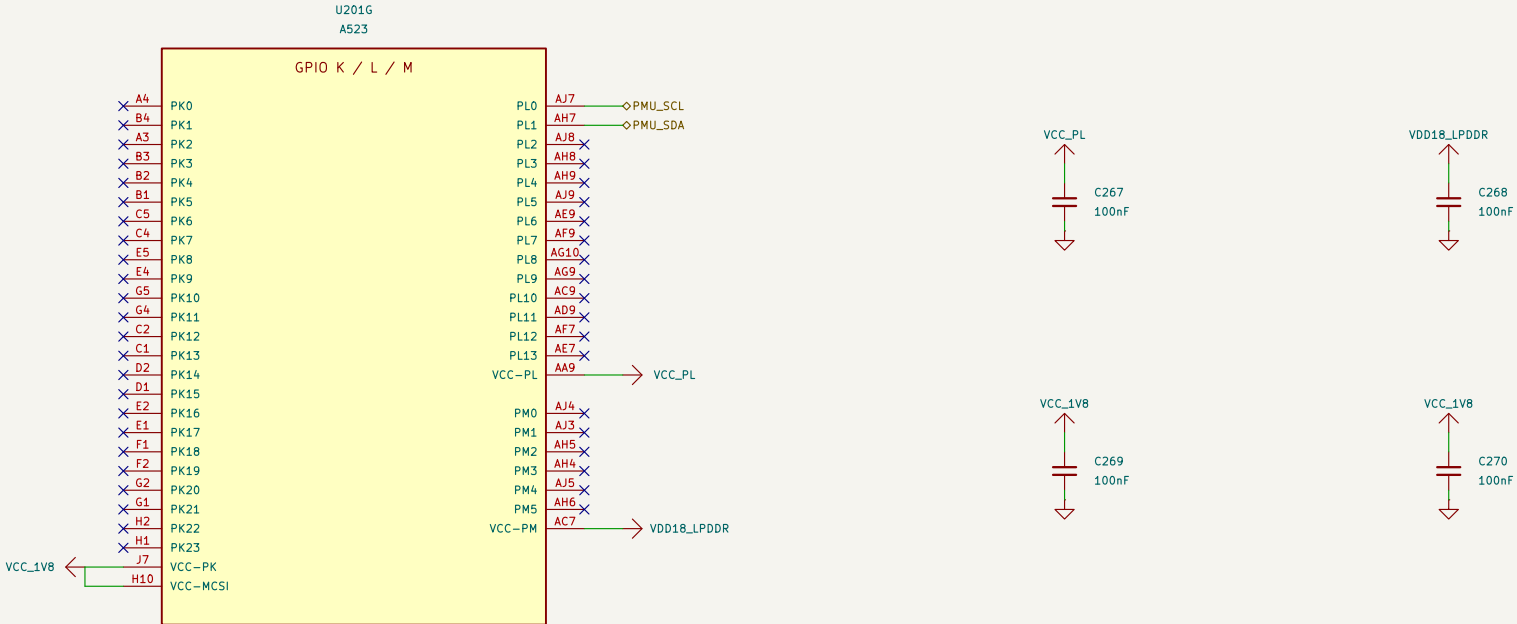


PD0..PD5: DSI0 mux 4. Clock = PD4/PD5.
PD6..PD9 unused with the two-lane panel.
Follow panel and A523 routing constraints.
PE camera bank remains disabled.

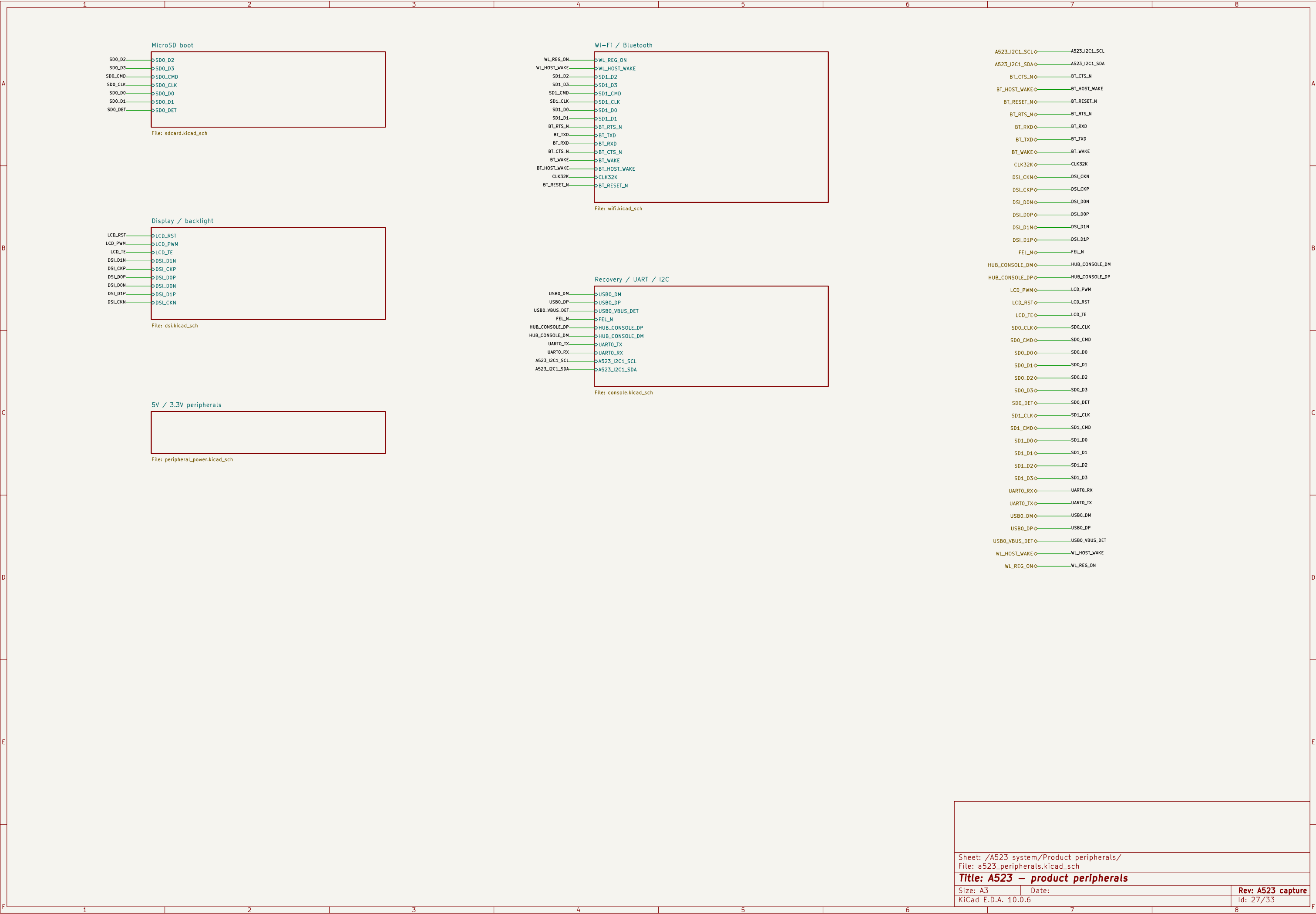
USB0 device/recovery / USB1 host

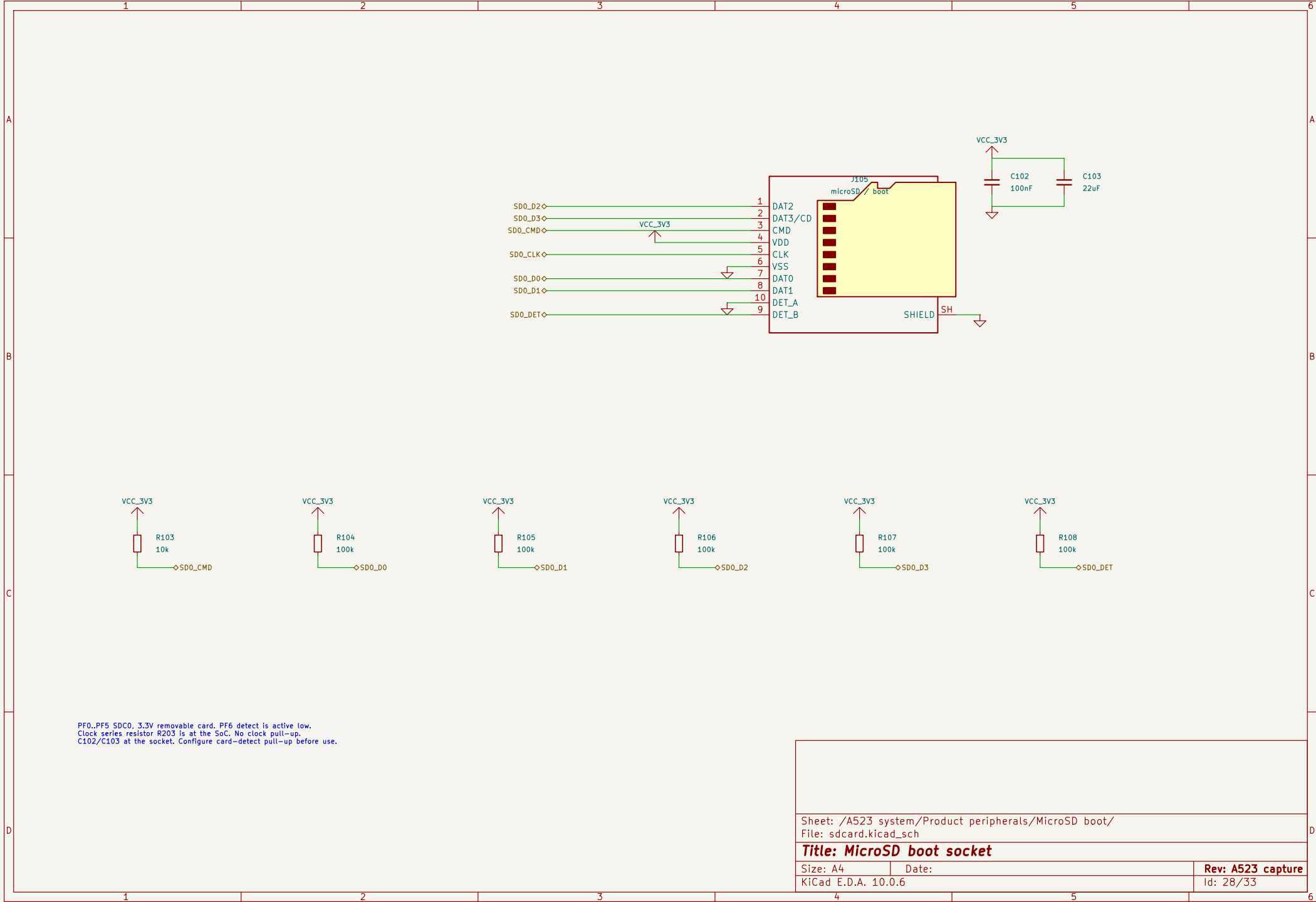


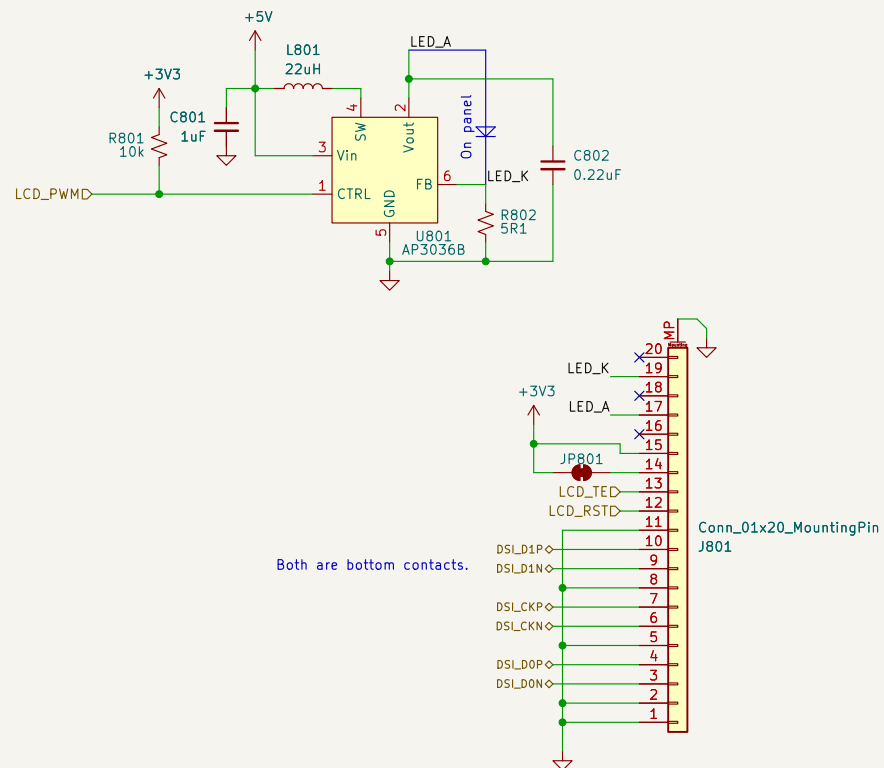
PMIC control bus and standby IO



PL0 / PL1: S-TWI0 mux 2, 3.3V.
PMIC addresses: 0x34 and 0x36.
MCSI must be supplied for SD voltage switching.
PK: unused, supplied at 1.8V.
PM: 1.8V standby; GPIOs unused.







Sheet: /A523 system/Product peripherals/Display / backlight/
File: dsi.kicad_sch

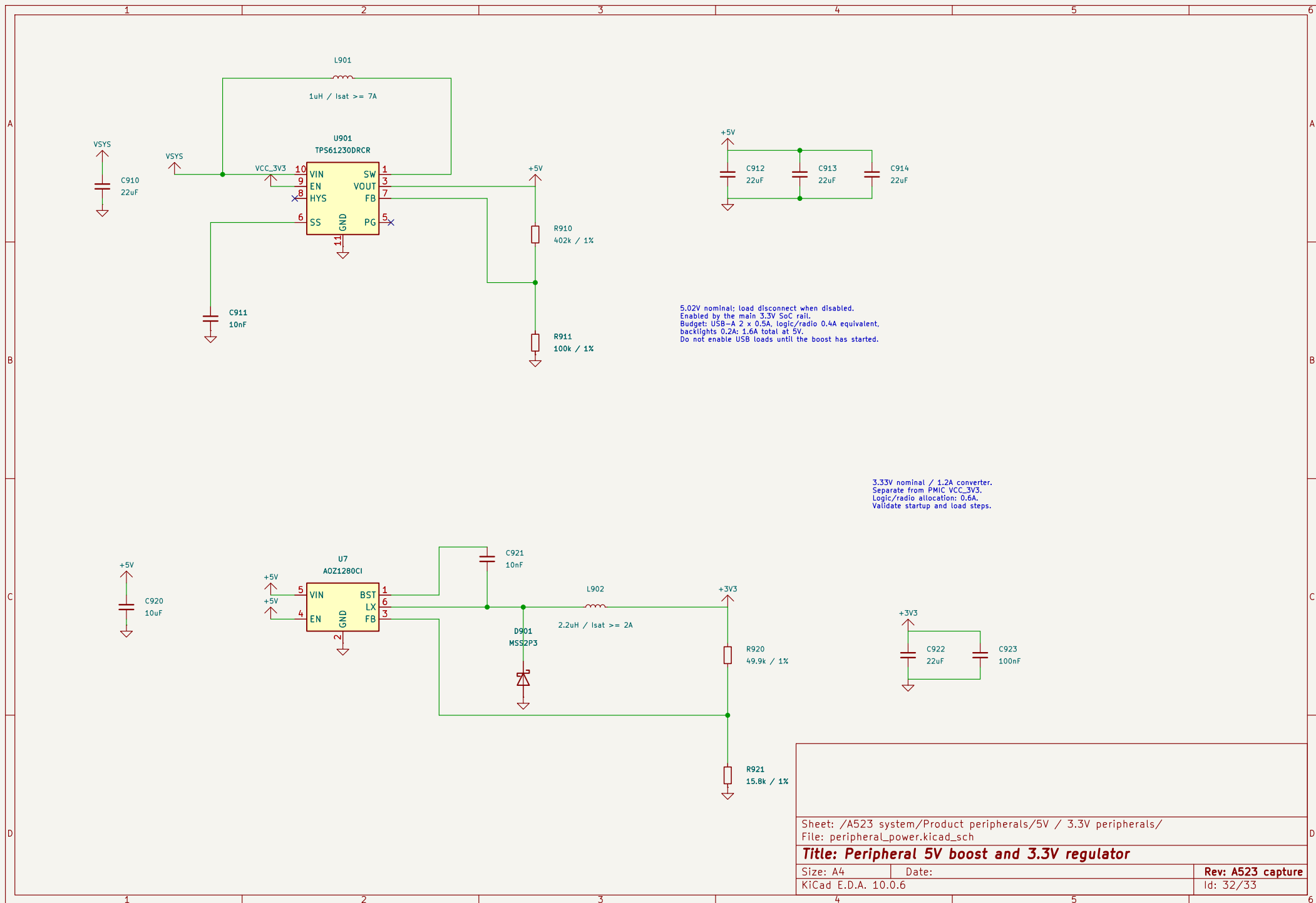
Title:

Size: A4 Date:

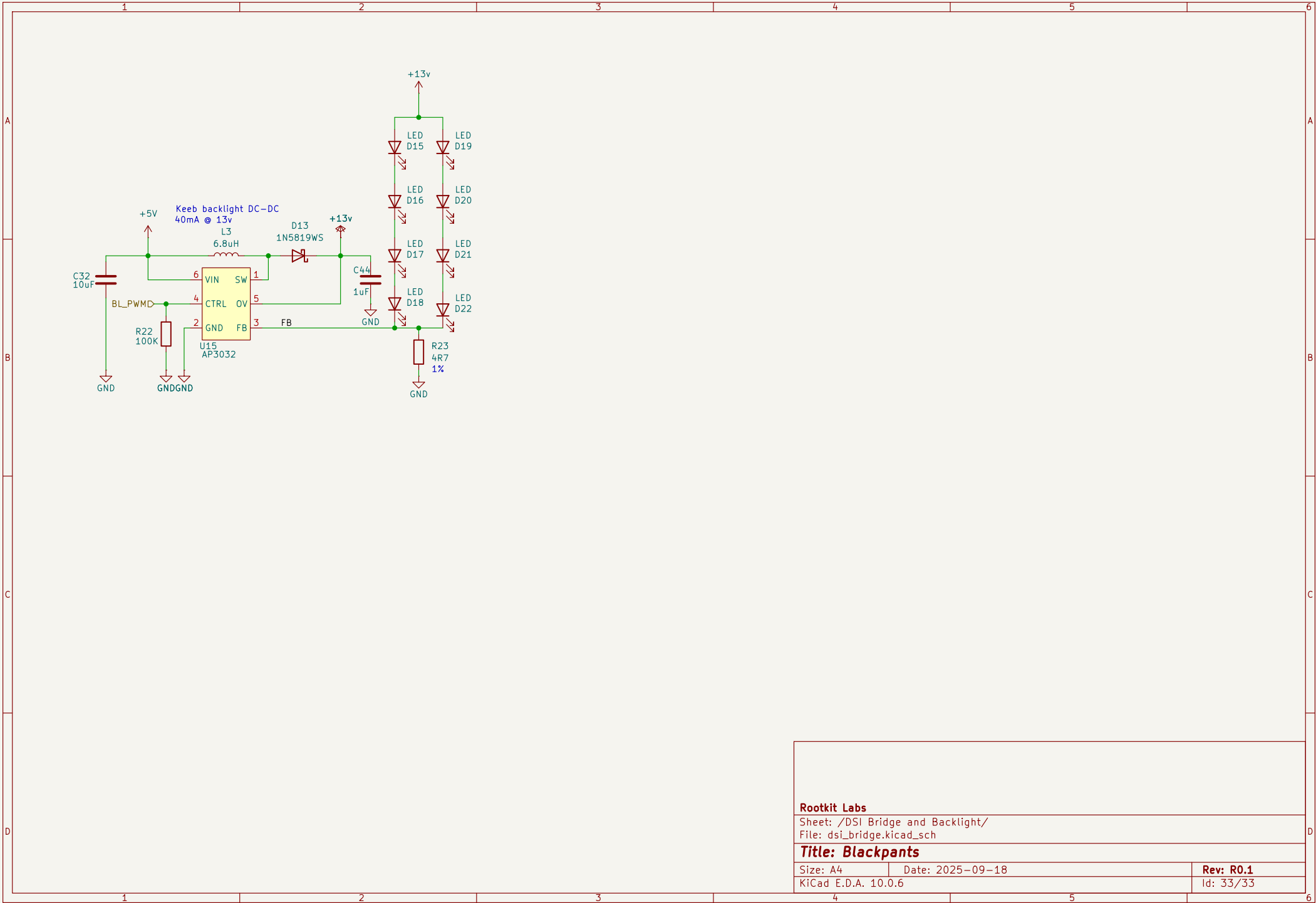
KiCad E.D.A. 10.0.6

Rev:

Id: 30/33



Sheet: /A523 system/Product peripherals/5V / 3.3V peripherals/ File: peripheral_power.kicad_sch		
Title: Peripheral 5V boost and 3.3V regulator		
Size: A4	Date:	Rev: A523 capture
KiCad E.D.A. 10.0.6		Id: 32/33



Rootkit Labs

Sheet: /DSI Bridge and Backlight/
File: dsi_bridge.kicad_sch

Title: Blackpants

Size: A4 Date: 2025-09-18
KiCad E.D.A. 10.0.6

Rev: R0.1
Id: 33/33